

SAROSWAT ROY

AI Engineer & Data Scientist · Agentic AI · NLP · Explainable AI · IT Operations

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PROFILE

AI Engineer and Data Scientist combining agentic systems, natural language processing, Generative AI, Explainable AI and real-world IT operations. Builds intelligent systems that are autonomous where appropriate, explainable where necessary, secure by design and operationally useful.

EXPERIENCE

Information Technology Support Engineer

City of London Corporation · City of London School · Sep 2025 – Present

Supports secure, reliable technology services and applies automation to operational workflows.

- Office 365 administration, technical troubleshooting, networking and device support.
- Cybersecurity, cloud-security and risk-aware incident handling.
- AI-assisted ticket triage, prioritisation and repetitive-work reduction.

Artificial Intelligence Research Graduate Assistant

City of London School · Sep 2024 – Aug 2025

Applied AI research and education with an emphasis on responsible use in learning environments.

- Machine learning, NLP, prompt engineering and AI-assisted teaching.
- Software testing, AWS experimentation and safe, age-appropriate AI support.

Research Assistant

University of Engineering & Management, Jaipur · Nov 2021 – Jun 2022

Contributed to research on semantic analysis of code.

- Code semantics, data modelling, statistical analysis and algorithms.
- Research methodology and technical documentation.

EDUCATION

MSc Artificial Intelligence — Natural Language Processing

Queen Mary University of London · Sep 2023 – Sep 2024 · Merit

Research: biomedical named entity recognition using transformer-based models. Study included NLP, NLU, deep learning, transformers, statistics, knowledge engineering and responsible AI.

BTech Computer Science

University of Engineering & Management, Jaipur · Aug 2019 – May 2023

Algorithms, data structures, programming, machine learning, software engineering, statistics and computer networks.

SELECTED ENGINEERING WORK

Agentic Guardrail Engine for Financial AI

[Next.js](#) · [FastAPI](#) · [SQLAlchemy](#) · [Docker](#) · [Governed agentic decision control](#)

Routes financial AI recommendations through deterministic policy, risk, fairness, audit and human-review controls.

- Versioned policies, explainable risk contributors and persisted overrides.
- Only approve, reject or escalate outcomes are permitted.

VeriNLI: Explainable Hallucination Verification

[Python](#) · [FastAPI](#) · [BM25](#) · [NLI](#) · [Evidence-grounded language systems](#)

Splits generated answers into claims, retrieves evidence, calibrates verdicts and produces reviewable reports.

- Handles contradiction, abstention and conflicting credible sources.
- Seven test modules cover the pipeline, API, retrieval, calibration and adversarial cases.

EduSentinel AI

[Python](#) · [FastAPI](#) · [SQLite](#) · [Responsible educational AI infrastructure](#)

Combines teaching support, safeguarding triage, policy intelligence, governance telemetry and IT operations in a local-first platform.

- Specialist service architecture with audit trails and human escalation.
- Includes policy grounding, explainability and multi-agent planning surfaces.

Biomedical Named Entity Recognition

[PyTorch](#) · [Hugging Face Transformers](#) · [Jupyter](#) · [MSc research](#)

Compared domain transformer families for biomedical token classification on JNLPBA.

- Strongest saved run: F1 0.7831, precision 0.7359, recall 0.8369.
- Confidence-aware demo can defer uncertain spans for human review.

QuantLab: LLM Quantisation Benchmarks

[NumPy](#) · [PyTorch](#) · [OpenVINO](#) · [FastAPI](#) · [React](#) · [Model efficiency](#)

Measures INT8 reconstruction, memory, perplexity, latency and throughput under reproducible settings.

- Separates simulated quantisation from deployable compressed inference.

CORE CAPABILITIES

AI systems: Agentic AI, AI agents, autonomous LLM systems, multi-agent workflows, Generative AI, prompt engineering, MCP, Ollama

NLP and ML: Transformers, biomedical NER, NLI, embeddings, RAG, BM25, PyTorch, TensorFlow, scikit-learn, graph neural networks, model evaluation

Responsible AI: Explainability, governance, guardrails, human-in-the-loop controls, calibration, risk management, data ethics and auditability

Engineering and operations: Python, TypeScript, React, Next.js, FastAPI, SQL, REST APIs, Docker, GitHub Actions, IT operations, Office 365, networking and cybersecurity

SELECTED PROFESSIONAL LEARNING

IBM Building AI Agents and Agentic Workflows · Duke Explainable AI · Oxford Saïd AI Governance · Google Cloud Responsible AI for Developers · IBM Data Science · IBM Cybersecurity Analyst · Google IT Support

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